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United States Patent
Kind Code
Date of Patent
Inventor(s)

12403280
B2
September 02, 2025
Hunt; Barry W. et al.

Apparatus, systems and method for collecting and reclaiming anaesthetic agents and for removing nitrous oxide from exhaust gases

Abstract

A system for collecting an anaesthetic agent, having at least one anaesthetic gas scavenging system (AGSS) for receiving exhaust gas from a plurality of sources, the exhaust gas including the anaesthetic agent to be collected, each AGSS comprising at least one power source for providing suction of the exhaust gas from the plurality of sources under negative pressure, and a central collection system for receiving the exhaust gas, the central collection system comprising at least one collector for collecting the anaesthetic agent from the exhaust gas, wherein the at least one collector is configured to adsorb the anaesthetic agent from the exhaust gas. The central collection system may be configured to receive the exhaust gases from the at least one AGSS, with the central collection system being located downstream of the at least one AGSS.

Inventors: Hunt; Barry W. (Cambridge, CA), Jarrett; Todd Eric (Cambridge, CA), Thorne; Damian Ross (Cambridge, CA), Ahchong; Katrina Marie (Cambridge, CA), Carr; Dean Paul (Cambridge, CA), Vasquez; Cesar Laurentino Martinez (Cambridge, CA)

Applicant: CLASS 1 INC. (Cambridge, CA)

Family ID: 44482424

Assignee: Class 1 Inc. (Cambridge, CA)

Appl. No.: 17/210943

Filed: March 24, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210205557 A1	Jul. 08, 2021

Related U.S. Application Data

continuation parent-doc US 15350576 20161114 ABANDONED child-doc US 17210943
continuation parent-doc US 13256367 US 9492628 20161115 WO PCT/CA2011/000181 20110217
child-doc US 15350576
us-provisional-application US 61306915 20100222
us-provisional-application US 61368608 20100728

Publication Classification

Int. Cl.: **B01D53/047** (20060101); **A61L9/20** (20060101); **A61M16/00** (20060101); **A61M16/01** (20060101); **A61M16/08** (20060101); **B01D53/26** (20060101); **B01D53/56** (20060101); **B01D53/75** (20060101); **B01J19/08** (20060101); **B01J20/02** (20060101); **G05D7/06** (20060101); **G06Q40/12** (20230101); **A61M16/10** (20060101); **B01D53/02** (20060101); **B01D53/22** (20060101)

U.S. Cl.:

CPC **A61M16/0093** (20140204); **A61L9/20** (20130101); **A61M16/009** (20130101); **A61M16/01** (20130101); **A61M16/024** (20170801); **A61M16/0808** (20130101); **B01D53/047** (20130101); **B01D53/261** (20130101); **B01D53/266** (20130101); **B01D53/268** (20130101); **B01D53/565** (20130101); **B01D53/75** (20130101); **B01J19/088** (20130101); **B01J20/02** (20130101); **G05D7/0623** (20130101); **G06Q40/123** (20131203); **A61L2209/14** (20130101); **A61M2016/0042** (20130101); **A61M16/0051** (20130101); **A61M16/0066** (20130101); **A61M16/101** (20140204); **A61M2202/0208** (20130101); **A61M2202/0241** (20130101); **A61M2202/0275** (20130101); **A61M2202/0283** (20130101); **A61M2202/062** (20130101); **A61M2205/3653** (20130101); **A61M2230/437** (20130101); **B01D53/02** (20130101); **B01D53/22** (20130101); **B01D53/56** (20130101); **B01D2257/102** (20130101); **B01D2257/104** (20130101); **B01D2257/40** (20130101); **B01D2257/504** (20130101); **B01D2259/4508** (20130101); **B01D2259/4533** (20130101); **Y02C20/10** (20130101); **Y02C20/40** (20200801); **Y10T137/0324** (20150401); **Y10T137/7722** (20150401); **Y10T137/7736** (20150401); **Y10T137/7761** (20150401)

Field of Classification Search

CPC: **A61M** (16/0093); **A61M** (16/024); **A61M** (16/009); **A61M** (16/01); **A61M** (16/0808); **A61M** (16/101); **A61M** (16/0051); **A61M** (16/0066); **A61M** (2016/0042); **A61M** (2202/0208); **A61M** (2202/0241); **A61M** (2202/0275); **A61M** (2202/0283); **A61M** (2202/062); **A61M** (2205/3653); **A61M** (2230/437); **G06Q** (40/123); **A61L** (9/20); **A61L** (2209/14); **B01D** (53/047); **B01D** (53/261); **B01D** (53/266); **B01D** (53/268); **B01D** (53/565); **B01D** (53/75); **B01D** (53/02); **B01D** (53/22); **B01D** (53/56); **B01D** (2257/102); **B01D** (2257/104); **B01D** (2257/40); **B01D** (2257/504); **B01D** (2259/4508); **B01D** (2259/4533); **B01J** (19/088); **B01J** (20/02); **G05D** (7/0623); **Y10T** (137/0324); **Y10T** (137/7722); **Y10T** (137/7736); **Y10T** (137/7761); **Y02C** (20/40); **Y02C** (20/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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5171553	12/1991	Li	423/239.2	B01D 53/8628
5231980	12/1992	Filipovic	128/205.12	A61M 16/0087
5366701	12/1993	Taylor	422/186.21	B01D 53/32
6347627	12/2001	Frankie	128/201.27	B01D 53/8625
6405539	12/2001	Stach	62/3.4	B01J 20/18
7235222	12/2006	Hotta	423/239.1	B01D 53/8631
9039808	12/2014	Filipovic	95/142	B01J 20/12
9492628	12/2015	Hunt	N/A	A61L 9/20
2005/0155380	12/2004	Rock	62/617	A61M 16/009
2012/0222556	12/2011	Filipovic	96/121	B01J 20/12
2021/0205557	12/2020	Hunt	N/A	B01D 53/266

Primary Examiner: McCalister; William M

Attorney, Agent or Firm: Smart & Biggar LP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 15/350,576 filed Nov. 14, 2016, which is a continuation of U.S. application Ser. No. 13/256,367 filed Nov. 13, 2011 (now U.S. Pat. No. 9,492,628 and which issued on Nov. 15, 2016), which was a U.S. National Stage Entry of PCT Application No. PCT/CA2011/000181 filed on Feb. 17, 2011, which claimed the benefit of U.S. Provisional Patent Application Ser. No. 61/306,915 filed on Feb. 22, 2010, and which also claimed the benefit of U.S. Provisional Patent Application Ser. No. 61/368,608 filed on Jul. 28, 2010, the entire contents of all applications being hereby incorporated by reference herein for all purposes.

TECHNICAL FIELD

(1) Embodiments described herein relate to apparatus, systems and methods for collecting and/or reclaiming anaesthetic agents, and in particular, for reclaiming halogenated drugs from exhaust gases such as waste air expelled by patients in operating rooms, as well as for removing other substances from exhaust gases such as nitrous oxide, oxygen, and biological substances.

INTRODUCTION

(2) Numerous devices are known for collecting and/or reclaiming anaesthetic agents such as halogenated drugs or nitrous oxide.

(3) For example U.S. Pat. No. 3,592,191 (Jackson) discloses a method and apparatus for recovering exhausted anaesthetic agents. Recovery is effected by collecting the exhausted gas and removing the water vapor either by condensation or with a hygroscopic material. The anesthetic agent is then extracted from the dried gas by a cryogenic process in which the vapors in the anesthetic agent are condensed to a liquid phase or by an absorbent material, which is processed later to remove the agent. When employing the condensing process, the anesthetic condensate is collected and may be readmitted directly into the anesthetic system.

(4) U.S. Pat. No. 7,235,222 (Hotta et al.) discloses a process and an apparatus for treating a waste anesthetic gas containing a volatile anesthetic and nitrous oxide discharged from an operating room by introducing the gas into an adsorbing cylinder filled with an adsorbent, where the volatile anesthetic contained in the waste anesthetic gas is adsorbed and thereby removed, and successively introducing the gas into a catalyst layer filled with a nitrous oxide decomposition catalyst, where nitrous oxide is decomposed into nitrogen and oxygen.

(5) U.S. Patent Application Publication Number 2005/0155380 (Rock) discloses devices for the

recovery of volatile, organic anesthetic agents from waste anesthesia gas. The device recovers the anesthetic agents by selectively condensing the agents in a cooling chamber and storing the condensed agents in a pressurized storage chamber.

(6) U.S. Patent Application Publication Number 2006/0254586 (Berry et al.) discloses a method and system for removal of nitrous oxide and volatile halocarbon gas components from waste anesthetic gases using a low-flow scavenging or reclamation system that preferably includes an intelligent waste anesthetic gas collection unit fluidly coupled between each individual anesthetic machine and the waste gas evacuation manifold. Through a system including a collection chamber, a pressure detector, and an exhaust valve which is actuated based on the detected pressure in the collection chamber, the waste anesthetic gas collection unit allows flow to the waste suction manifold only in the presence of waste gas and interrupts all flow into the suction manifold when no waste gas is present.

(7) U.S. Patent Application Publication Number 2006/0254589 (Berry et al.) discloses a method and apparatus for recovering and separating anesthetic gas components from waste anesthetic gases to be purged from a healthcare facility. Prior to a condensation step, a compressor is used to increase the waste anesthetic gas pressure in order to facilitate condensation of anesthetic gas components at higher temperatures and in greater amounts than through condensation at lower pressures. Condensing the anesthetic gas components from the compressed waste anesthetic gas stream is then achieved using conventional condensation systems, which remove anesthetic gases as either liquid condensates or solid frosts. Some embodiments may be used with existing high-flow scavenging or reclamation systems but are more preferably used with low-flow scavenging or reclamation systems, which employ intelligent waste anesthetic gas collection units to minimize the ingress of atmospheric gas when no waste anesthetic gas is to be purged from the healthcare facility.

(8) In spite of these known devices, a need for apparatus, systems and methods of collecting and/or reclaiming anaesthetic agents such as halogenated drugs or nitrous oxide has been recognized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings included herewith are for illustrating various examples of methods and apparatus of the present specification and are not intended to limit the scope of what is taught in any way. In the drawings:

(2) FIG. 1 is a schematic representation of a collection system for collecting anaesthetic agents according to one embodiment;

(3) FIG. 2 is a schematic representation of a monitoring system for use with the collection system of FIG. 1;

(4) FIG. 2A is a flow chart illustrating a method for collection of anaesthetic agents according to another embodiment;

(5) FIG. 3 is a schematic representation of an operating room and a conservation valve of the collection system of FIG. 1;

(6) FIG. 4 is a schematic representation of the conservation valve of FIG. 3;

(7) FIG. 5 is a schematic representation of a conservation valve connected to an anaesthetic machine according to another embodiment;

(8) FIG. 6 is a schematic representation of a first subsystem of a collection system for collecting anaesthetic agents;

(9) FIG. 7 is a schematic representation of another subsystem for use with the first subsystem of FIG. 6, and which is operable to remove nitrous oxide from exhaust gases using a reaction chamber;

- (10) FIG. **8** is a schematic representation of another subsystem for use with the first subsystem of FIG. **6**, and which is operable to remove nitrous oxide from exhaust gases using a reaction chamber;
- (11) FIG. **9** is a schematic representation of another subsystem for use with the first subsystem of FIG. **6**, and which is operable to remove and collect nitrous oxide;
- (12) FIG. **10** a schematic representation of another subsystem for use with the first subsystem of FIG. **6**, and which is operable to remove nitrous oxide and oxygen from exhaust gases;
- (13) FIG. **11** is a schematic representation of a first subsystem of a yet another collection system for collecting anaesthetic agents, and which is operable to remove biological substances;
- (14) FIG. **12** a schematic representation of another subsystem for use with the first subsystem of FIG. **11**, and which is operable to disinfect exhaust gases prior to collecting the anaesthetic agents;
- (15) FIG. **13** is a schematic representation of a first subsystem of a yet another collection system for collecting anaesthetic agents; and
- (16) FIG. **14** a schematic representation of another subsystem for use with the first subsystem of FIG. **13**, and which is operable to disinfect exhaust gases prior to collecting the anaesthetic agents.
- DETAILED DESCRIPTION**

- (17) Referring to FIG. **1**, illustrated therein is a collection system **10** for collecting or reclaiming anesthetic agents such as halogenated drugs or nitrous oxide from one or more sources of exhaust gas according to some embodiments. The collection system **10** may be installed in a healthcare facility such as a hospital and may be centrally located such that it is in fluid communication with one or more sources of exhaust gas, such as one or more operating rooms **12**, **14**, and **16**.
- (18) Furthermore, the collection system **10** may be remotely located relative to the operating rooms **12**, **14**, **16** and other sources of exhaust gas. For example, the collection system **10** may be located within the hospital at a location that is both central and remote relative to the individual operating rooms **12**, **14**, **16** while remaining in fluid communication with the operating rooms **12**, **14**, **16**, (e.g. via piping, ducting or other mediums for transporting liquids or gases, collectively referred to as “piping” herein).
- (19) The operating rooms **12**, **14**, **16** may be the source of one or more exhaust gases. For example, each operating room **12**, **14**, **16** may have an anaesthetic machine **20** that can be connected to one or more patients **18** for administering one or more anaesthetic agents (such as halogenated drugs, nitrous oxide, etc.) during or in association with a medical procedure.
- (20) In some cases, the anaesthetic machine **20** may also collect exhaust gases from the patient **18** and direct those exhaust gases to the collection system **10**, for example, through an exhaust port **24** in the operating room. In some embodiments, a conservation valve **22** may be located between the anaesthetic machine **20** and the exhaust port **24** as will be described below.
- (21) In some embodiments, the exhaust gases may come from other sources, such as an outpatient clinic, a surgery clinic, a doctor's office, an oral surgery clinic, a veterinary clinic, or other types of healthcare facilities.
- (22) As shown, an Anaesthetic Gas Scavenging System **30** (“AGSS”), also known as a Waste Anaesthetic Gas Disposal (“WAGD”), may be connected to the operating rooms **12**, **14**, **16** and the collection system **10**. As shown, the AGSS **30** may be located between the operating rooms **12**, **14**, **16** and the collection system **10**.
- (23) Generally, the AGSS **30** draws exhaust gases from the operating rooms **12**, **14**, **16** or another source, and directs these exhaust gases to the collection system **10**. For example, the AGSS **30** may include a power source such as a vacuum pump, blower, or fan connected to the exhaust ports **24** of the operating rooms **12**, **14**, and **16** through piping. In some instances, the power source may be connected to an inlet port **32** of the collection system **10** through piping.
- (24) In some embodiments, the AGSS **30** may include two or more power sources connected in parallel. Having an additional power source may provide a back-up in case one power source stops operating (e.g. breaks down or needs maintenance), which may improve system redundancy.

Additional power sources may also increase the suction, for example, when the AGSS 30 is connected to larger systems (e.g. a greater number of operating rooms). Furthermore, in some embodiments there may be more than one AGSS 30, which may be connected in parallel, for example, to provide redundancy.

(25) Generally, the collection system 10 includes one or more collectors 42 for collecting and removing anesthetic agents from the exhaust gas. The collection system 10 may also include a dryer 40 for removing moisture from the exhaust gas, and may include a dryer regeneration module 60 operatively coupled to the dryer 40 as will be described below.

(26) The collection system 10 may also include a controller 80 for controlling various subcomponents of the collection system 10 such as the collectors 42, the dryer 40 and the dryer regeneration module 60 as will be described below.

(27) Under normal operating conditions, exhaust gases entering the collection system 10 typically pass through the dryer 40 first, and then through the collector 42. Accordingly, the dryer 40 and the collector 42 may be in fluid communication with each other.

(28) After flowing through the dryer 40 and the collector 42, the exhaust gas may exit the collection system 10 at an outlet port 44, which may be coupled to an exhaust 46 that vents to atmosphere outside of a medical facility.

(29) As shown, the dryer 40 is in fluid communication with the inlet port 32 and receives exhaust gas from the AGSS 30. Often, the exhaust gas collected from the operating rooms 12, 14, and 16 may include moisture entrained in the gas flow, which may adversely affect the amount of anesthetic agent collected by the collectors 42. For example, moisture may be exhaled by the patients 18 in the operating rooms 12, 14, 16 during medical procedures.

(30) The dryer 40 helps to remove this moisture from the exhaust gas. For example, the dryer 40 may include desiccant material such as activated alumina, or may use other methods of drying air, such as condensation or membrane dryers, as will be described below.

(31) The dryer 40 may be any suitable shape and size. For example, the dryer 40 may be a tower having an inlet at the bottom of the tower and an outlet at the top of the tower. In some cases, the size and shape of the tower may be determined according to the amount of moisture to be removed.

(32) In one example, the dryer 40 may include approximately 2.5 kilograms of alumina desiccant for each operating room (e.g. operating rooms 12, 14, 16), which may allow the dryer 40 to continuously dehumidify exhaust gases from a typical patient for approximately eight hours. For example, in the illustrated embodiment, there are three operating rooms 12, 14, 16, and the dryer 40 may include at least 7.5 kilograms of alumina desiccant.

(33) In some embodiments, more activated alumina may be provided in the dryer 40 so as to provide a buffer. For example, in the illustrated embodiment, approximately 10 kilograms of activated alumina may be provided in the dryer 40. The dryer 40 may also include more desiccant so as to provide a longer operating time (e.g. sixteen hours of continuous operation).

(34) After flowing through the dryer 40, the dryer 40 outputs a dehumidified exhaust gas, which is then received by one or more collectors 42. In the illustrated embodiment, the collectors 42 adsorb anaesthetic agents from the exhaust gas. For example, the collectors 42 may include hollow vessels containing adsorbent material made from hydrophobic molecular sieves as generally described in U.S. Pat. No. 5,231,980 (Filipovic). The hydrophobic molecular sieves may be of the high silica zeolite type. According to Filipovic, such adsorbent materials have pore diameters large enough to permit molecules of halogenated drugs to pass therethrough and be selectively adsorbed in large internal cavities of the crystal framework, whereby halogenated drugs may be removed.

(35) While the embodiment described above refers to a hydrophobic molecular sieve for collecting and reclaiming halogenated drugs, the collector 42 may collect anaesthetic agents using different materials or methods. For example, the collector 42 may utilize other adsorbent materials such as other types of molecular sieves or activated charcoal to collect and reclaim halogenated drugs. In some cases the collector 42 could use a cryogenic collector.

(36) In normal operation, the exhaust gas passes through the collector **42** even when the collector is saturated or cannot remove anaesthetic agents from the exhaust gas flow. However, once the collector is full, the flow of exhaust gases may be diverted from the collector **42** that is full, and the anaesthetic agents may be recovered from that particular collector **42** for later reuse.

(37) For example, when using the hydrophobic molecular sieve described by Filipovic, the exhaust gases may pass through a bed of adsorbent material until the material in the sieve is saturated to the extent that breakthrough of the hydrocarbons is determined (e.g. halogenated hydrocarbons are detected at the outlet of the collector). Next, the adsorbent material with the adsorbed phase of halogenated hydrocarbons may be removed from the system and regenerated by exposing the saturated material to an inert purging gas stream under conditions which desorb the halogenated hydrocarbons from the adsorbent material into the purging gas stream. The halogenated hydrocarbons may then be removed from the purging gas stream and purified to a desired level of purity for reuse of the recovered halogenated hydrocarbons.

(38) In some embodiments, a third party may remove the anaesthetic agents (e.g. halogenated hydrocarbons) from the collector **42**.

(39) In some embodiment, after recovering the anaesthetic agents from the collector **42**, the collector **42** may be reinstalled in the system **10**. Accordingly, the collector **42** may be reused multiple times.

(40) When recovering anaesthetic agents from the collector **42** it may be desirable to keep the system **10** operational so as to provide continuous collection of anaesthetic agents. Accordingly, the system **10** may include a plurality of collectors **42** (e.g. collector bank "A" and collector bank "B") connected together in parallel so that at least one collector **42** remains operational while one or more other collectors **42** are being processed to remove the captured anaesthetic agents, for example.

(41) When there is a plurality of collectors **42** connected in parallel, the inlet of each collector **42** may include a valve **50** for selectively opening and closing the flow of exhaust gases into the respective collector **42** from the dryer **40**. This valve **50** may be closed when removing the respective collector **42** from the system **10**, for example, when recovering anaesthetic agents from each respective collector **42**. At the same time, other collectors **42** may have open valves **50** such that dehumidified exhaust gases are permitted to flow into those other collectors. Accordingly, the collection system **10** may remain operational while some of the collectors **42** are being processed.

(42) The collection system **10** may also include a sensor **52** for measuring an amount of anaesthetic agent collected and for detecting when one of the collectors **42** is full, for example, so that anaesthetic agents can be reclaimed, for example, by removing/exchanging the collector **42** from the system **10**. In some embodiments, the sensor **52** may be a weight scale that measures the weight of the collector **42** (e.g. when the collector **42** is a hollow vessel). Upon reaching a particular weight, the collector **42** may be removed/exchanged from the system **10** so that the anaesthetic agent may be recovered from the collector **42**.

(43) In some embodiments, the collector **42** may have multiple stages for removing anaesthetic agents from the exhaust gas. For example, each collector **42** may include a first stage **54** and a second stage **56**. The stages **54**, **56** may use one or more different methods or devices for collecting or reclaiming anaesthetic agents from the exhaust gas. For example, the first stage **54** may include a first type of adsorbent material such as a molecular sieve that tends to adsorb a first type of anaesthetic agent (e.g. Sevoflurane or Desflurane, or both) for later reuse. The second stage **56** may include a second adsorbent material such as a molecular sieve that tends to adsorb a second type of anaesthetic agent (e.g. a halogenated drug such as Isoflurane) for later reuse.

(44) Accordingly, a multi-stage collector **42** may allow the collection system **10** to collect two or more types of anaesthetic agents for later reuse. This may be useful when implementing a centralized collection system **10** that recovers anaesthetic agents from multiple sources of exhaust gas that may contain multiple different anaesthetic agents.

(45) In some embodiments, each of the first and second stages **54**, **56** may collect more than one type of anaesthetic agent. For example, the first stage **54** may collect a plurality of anaesthetic agents such as all halogenated drugs. However, even if the first stage collects a plurality of anaesthetic agents, the first stage **54** may preferentially collect one or more anaesthetic agents in comparison to other anaesthetic agents. For example, the first stage **54** may collect all halogenated drugs, but may also collect Sevoflurane and Desflurane to a greater extent than Isoflurane.

(46) While the illustrated embodiment shows two stages **54**, **56**, the collector **42** may include any number of stages. For example, there may be additional stages to reclaim other anaesthetic agents for later reuse, including but not limited to other halogenated drugs or nitrous oxide. In some embodiments, the collector **42** may include a single stage.

(47) In some embodiments, the multiple stages of the collector **42** may serve purposes other than reclaiming multiple types of anaesthetic agents for later reuse. For example, the second stage **56** may include a second adsorbent material, such as activated charcoal, that functions as a fail-safe that may help prevent at least some anaesthetic agents from exiting the collection system **10**, and which could otherwise lead to pollution.

(48) For example, activated charcoal may adsorb one or more anaesthetic agents, but it may be impractical to recover those adsorbed anaesthetic agents from the activated charcoal. In this sense, the activated charcoal may be disposable and may merely sequester the anaesthetic agents, as opposed to allowing them to be reclaimed for later reuse. Nonetheless, providing a second stage **56** with activated charcoal or similar materials may help reduce pollution and generally provide for cleaner exhaust gases.

(49) In some embodiments, one or more collectors **42** or stages thereof may act as a sequestration device that collects anaesthetic agents for disposal, as opposed to later reuse. For example, the collector **42** may be a vessel containing activated charcoal and it may be impractical to recover the anaesthetic agents from the activated charcoal for reuse.

(50) In some embodiments, the collection system **10** may not include a dryer **40**. For example, this may be possible when the system **10** includes a collector **42** that functions relatively independent of moisture content in the exhaust gas flow, such as a molecular sieve (which could be hydrophobic). However, providing a dryer **40** may improve the efficiency of a hydrophobic molecular sieve, for example, because it has been determined that hydrophobic molecule sieves might tend to adsorb some water in addition to anaesthetic agents.

(51) When the system **10** includes a dryer **40**, it may become saturated or otherwise unable to remove moisture from the exhaust gas. Accordingly, the collection system **10** may include a dryer regeneration module **60** for removing or purging collected moisture from the dryer **40**. For example, when the dryer **40** includes alumina desiccant, the regeneration module **60** may be operable to blow air through the desiccant in the dryer **40** under proper conditions that purge moisture from the dryer **40** and recharge the desiccant.

(52) For example, the regeneration module **60** may include an inlet filter **62**, a blower **64**, a check valve **66** and a heater **68**. In use, the blower **64** draws a purging gas such as air through the inlet filter to remove impurities that may otherwise damage the desiccant in the dryer **40**. The filtered air exits the blower **64** and passes through a one-way check-valve **66** and then into the heater **68** where the air is heated to a temperature sufficient to remove moisture from the desiccant in the dryer **40**. The air then proceeds through the dryer **40** and may then vent to atmosphere through the exhaust **46**.

(53) As described above, the heater **68** heats the air to a temperature sufficient to purge moisture from the desiccant. For example, the temperature may range from approximately 60° C. to 300° C. In some embodiments, the regeneration module **60** may provide air at approximately 200° C. and 20 cubic feet per minute for approximately six hours, which may be sufficient to remove moisture from approximately 7.5 kilograms of activated alumina desiccant.

(54) In some cases, the dryer **40** is generally allowed to cool after the desiccant is purged with hot

air. The cooling time may be approximately two hours. After this time, the dryer **40** may be used once again to remove moisture.

(55) As described above, the dryer **40** may be configured to operate continuously for between about 8-16 hours after being regenerated by the dryer regeneration module **60**, which may be sufficient to operate the collection system **10** for an entire day. At the end of the day, the dryer **40** may be purged of collected moisture using the process described above. This process may take approximately 8 hours, which means in some cases the system **10** can be purged at night while the system **10** is not in use. Once purged, the system **10** can resume normal operations the following day.

(56) In some embodiments, the dryer **40** may be configured to operate for longer periods of time. In some embodiments, there may be two or more dryers **40** fluidly coupled in parallel so that the system **10** can operate continuously. In this sense, the dryers **40** may have a similar configuration to the collectors **42** described above.

(57) When purging moisture from the dryer **40** using the regeneration module **60**, the air may proceed through the dryer **40** in reverse (e.g. from the outlet of the dryer to the inlet of the dryer). For example, the regeneration module **60** may be connected to an outlet port located near the top of the dryer **40**, and an inlet port on the bottom of the dryer **40** may be connected to the exhaust **46**. This configuration may help purge moisture from the dryer **40** because with high down flow purge air the desiccant material may be compacted and may increase the surface area contact, which may tend to avoid channelling, fluidization and dust release from the desiccant bed.

(58) In some other embodiments, the regeneration module **60** may blow air into the dryer **40** in the normal flow direction.

(59) While the illustrated embodiment shows the dryer **40** having two inlet ports and two outlet ports, there may be one inlet port and one outlet port that are connected to respective components, for example, using T-connectors or manifolds.

(60) In some embodiments, the system **10** may include control valves **50**, **70**, and **72** that control the direction of airflow through the dryer **40** and other elements of the system **10**. These valves **50**, **70** and **72** may inhibit air from flowing into the AGSS **30** or into the collectors **42** when the regeneration module **60** blows air through the dryer **40**. For example, when regenerating the dryer **40**, valves **50** and **72** may be closed, and valve **70** may be open. The outlet of the valve **70** may be connected to the outlet port **44** of the system **10** for venting air to atmosphere. Once the dryer **40** has been purged and regenerated, the valve **70** may be closed and the other valves **50** and **72** may be opened to resume normal operations and continue collecting or reclaiming anaesthetic agents.

(61) In some embodiments the valves **50**, **70**, **72** may be manually controlled.

(62) In some embodiments, the valves **50**, **70**, **72** may operate automatically. For example, the valve **72** may be a check valve. In another example, the valves may be controlled by a controller **80** or monitoring system **90** as will be described below.

(63) In some embodiments, the controller **80** (such as a control panel) may be configured to automate the system **10**. For example, the controller **80** may monitor the status of the dryer **40** and/or the status of the collectors **42** (e.g. via the sensor **52**) to determine if they are full and need to be replaced or regenerated. The controller **80** may also operate the dryer regeneration module **60** and the valves **50**, **70** and **72** depending on the capacity of the dryer **40**. For example, there may be a sensor (e.g. similar to or the same as sensor **52**) for measuring the amount of moisture collected by the dryer **40** and the controller **80** may be in communication with that sensor, for example, to operate the dryer regeneration module **60** when the dryer **40** is full or almost full.

(64) In some embodiments, the system **10** may be manually operated. For example, the dryer regeneration module **60** and the valves **50**, **70**, **72** may be manually operated. The system **10** may also have both automated and manual operation modes, for example, to provide system redundancy and safety.

(65) In some embodiments, the system **10** may include a bypass or failsafe so that exhaust gases

may bypass the dryer **40** and/or collectors **42** and/or other components of the system **10**. It may be desirable to use a bypass when conducting maintenance or servicing of the system **10**.

(66) For example, the collection system **10** may include a bypass valve **84** connected between the inlet port **32** and the outlet port **44**. In some embodiments, the bypass valve **84** may be operated automatically. In this sense, the bypass valve **84** may be a fail-safe valve that enhances safety, such as a normally open, energized closed, spring return ball valve, piston valve, blow-off valve or a burst disc. In other embodiments, the bypass valve **84** may be operated manually.

(67) The system may also include control valves **86**, **88** that isolate the components of the system **10**, for example, when the bypass valve **84** is in use. For example, valves **86**, **88** may be closed when the bypass valve **84** is open. The opening and closing of the valves **84**, **86**, **88** may be automated and may occur at the same time or at different times.

(68) As described above, the collection system **10** may be located centrally and remotely from one or more sources of exhaust gas (e.g. the operating rooms **12**, **14** and **16**). Having a centrally located system may decrease the overall cost of the collecting or reclaiming anaesthetic agents in comparison to conventional systems that are located at or near the source of the exhaust gas. As such, the initial capital costs of providing a large central collection system may be less than the accumulative cost of several smaller localized systems.

(69) Furthermore, operational costs of a centralized system may be lower in comparison to a local system because the centralized system **10** may be able to more easily implement a monitoring system that controls the collection and reclamation of anaesthetic agents from each source of exhaust gas. Tracking the collection of multiple sources of exhaust gas may help determine when servicing or maintenance is necessary, which may result in increased uptimes. For example, the monitoring system may track the capacity of the dryer **40** and/or collectors **42** so that they may be replaced or regenerated once they are saturated or otherwise full. Monitoring one central system in this way may be less expensive than monitoring several individual localized systems.

(70) Referring now to FIG. 2, a sample monitoring system **90** will now be described in greater detail. The monitoring system **90** generally includes a processor **92** that is in electronic communication (e.g. via a wired or wireless network) with the collection system **10** and the various subcomponents thereof such as the dryer **40**, the collectors **42**, the valves **50**, **70**, **72**, the sensors **52**, and the regeneration module **60**.

(71) The processor **92** may be a computer, microprocessor or another other suitable device for monitoring the collection system **10**. In some embodiments, the processor **92** may be implemented in electronics that are part of the controller **80** within the system **10**. In some embodiments, the processor **92** may be located separately and remotely from the collection system **10** altogether, and may be in communication with a plurality of collection systems **10** as will be described in greater detail below. In these cases, the processor **92** may be in communication with the controller **80** of each of the plurality of collection systems **10**.

(72) Generally, the processor **92** may communicate with the sensors **52** to determine when the collectors **42** are full. When the collectors **42** are full (or when the sensor **52** measures a particular amount of anaesthetic agent collected), the monitoring system **90** may initiate reclamation of the anaesthetic agent, for example, by initiating a process to purge or exchange the collector **42** that is full. For example, when using some collectors **42** with hydrophobic molecular sieves, it may be necessary to remove and/or exchange the collector **42** (e.g. a hollow vessel of hydrophobic material) from the system **10** to recover the anaesthetic agents captured therein. When one collector **42** is being removed/exchanged, the processor **92** may divert the flow of exhaust gases to one or more other collectors **42** that are still operational, for example, by opening or closing the valves **50**.

(73) Furthermore, when one or more hollow vessels are full, the monitoring system **90** may alert a collection agent **94** in order to request pick up and exchange the full vessels with empty ones. For example, the processor **92** may communicate with a third party collection agent **94** via a communications network **96** such as the Internet. After receiving the alert, the third party collection

agent **94** may recover anaesthetic agents from the full vessels and may sell the anaesthetic agents back to pharmaceutical companies **99**, directly to the medical facility, and so on. In some embodiments, the collection agent **94** may be an affiliate of the entity that provides the monitoring system **10** or monitoring services.

(74) The monitoring system **90** may also be adapted to keep track of the overall amount of anaesthetic agents recovered for tax saving or other accounting purposes. For example, the amount of anaesthetic agents recovered by a particular legal entity may allow that entity to claim tax credits such as carbon credits. Being able to track these tax credits may be particularly beneficial to drug companies and hospitals and may provide financial benefits.

(75) To keep track of this tax saving or other accounting information, the monitoring system **90** may include a data storage medium **98** such as a hard drive or database connected to the processor **92**. The processor **92** may keep track of the overall amount of anaesthetic agents recovered and the storage medium **98** may store the tax savings information for later use.

(76) The monitoring system **90** may also track the status of multiple collection systems **10**. For example, there may be a plurality of medical facilities in a particular geographic area or region, with each medical facility having a centralized collection system **10**. The processor **92** may be in communication with each of these collection systems **10** directly or through the communications network **96** so as to track overall collection and reclamation of anaesthetic agents for that geographic region. This information may be stored on the storage medium **98** for later use (e.g. for calculating tax credits).

(77) In some embodiments, the monitoring system **90** may be adapted to provide supply chain management for a particular geographic region, for example, by keeping track of the demand of each healthcare facility and the amount of anaesthetic agents each healthcare facility recovers. The monitoring system **90** may then allocate the recovered anaesthetic agents to the facilities based on demand, and may order fresh anaesthetic agents (e.g. from a pharmaceutical supply company) based on demand. Accordingly, in such embodiments the processor **90** may be in communication with the pharmaceutical company **99**, for example, through the communications network **96**.

(78) Referring now to FIG. 2A, illustrated therein is a method **100** for monitoring collection of anaesthetic agents.

(79) Step **101** includes communicating with at least one collection system. For example, the processor **92** may communicate with the one or more collections systems over a wired or wireless communication network. The collection system(s) may be similar to the collection system **10** described above. Each collection system generally includes at least one collector (e.g. the collector **42**) for collecting an anaesthetic agent from an exhaust gas that includes the anaesthetic agent. In some embodiments, step **101** may include communicating with a plurality of collection systems, which may be located remotely.

(80) Step **102** includes measuring an amount of anaesthetic agent collected by each collector. For example, the sensors **52** may be used to measure the amount of anaesthetic agent collected by the collectors **42**.

(81) Step **103** includes initiating reclamation of the anaesthetic agent from a collector after measuring a particular amount of anaesthetic agent collected by the respective collector. For example, reclamation may be initiated when the sensor **52** detects that the respective collector **42** is full, or almost full.

(82) Step **104** includes tracking an overall amount of anaesthetic agent collected. For example, the storage medium **98** may be used to track the overall amount of anaesthetic agent collected. Furthermore, the storage medium **98** may track the overall amount of anaesthetic agent collected by a plurality of collection systems.

(83) Step **105** includes calculating tax credits corresponding to the overall amount of anaesthetic agent collected. For example, the processor **92** may calculate the tax credits.

(84) Referring again to FIG. 1, as shown the collection system **10** is located downstream from the

AGSS **30**. Furthermore, in the illustrated embodiment, the AGSS **30** is an active system, which tends to provide a positive pressure downstream of the AGSS **30** (e.g. approximately 3 psi) and provides suction of the exhaust gas under negative pressure upstream of the AGSS **30** (e.g. approximately 12-20 inches-Hg). Connecting the collection system **10** to the exhaust side of the AGSS **30** provides an opportunity to collect anaesthetic agents under positive pressure, which may increase the efficiency of the collectors **42** because the adsorption of anaesthetic agents may be reduced under high negative pressures (e.g. 12-20 inches-Hg).

(85) In some embodiments, the AGSS **30** may include a vacuum pump and the collection system **10** may be connected to the outlet of the vacuum pump. Accordingly, the collection system **10** might receive exhaust gases having a positive pressure, and may thereby collect aesthetic agents under positive pressure, which may increase the efficiency of the collection system **10**, particularly when utilizing adsorption collectors **42** that include molecular sieves.

(86) In some other embodiments, the collection system **10** may be located upstream of the AGSS **30**. For example, the collection system **10** may be located upstream of a passive AGSS, which may provide a low negative pressure, such as approximately 0.1 inches-WC (water column).

(87) Referring now to FIGS. **1**, **3** and **4**, the conservation valve **22** will now be described in greater detail.

(88) As shown in FIGS. **1** and **3**, the conservation valve **22** is located between the AGSS **30** and the anaesthetic machine **20**. The conservation valve **22** is operable to selectively open and close the fluid passage between the anaesthetic machine **20** and the AGSS **30**, for example, depending on whether or not the anaesthetic machine **20** is actually in use. This may increase the efficiency of the collection system **10** and/or AGSS **30** as will be described below.

(89) Referring now to FIG. **4**, as shown the conservation valve **22** includes a valve body **108** having an inlet port **112**, an outlet port **114**, and a fluid passageway **110** therebetween. The conservation valve **22** also includes an actuator **116** for selectively opening and closing the fluid passageway **110** between the inlet **112** and outlet **114**.

(90) Generally, the inlet port **112** is connected to a source of exhaust gas (e.g. via the anaesthetic machine **20**) so as to allow the exhaust gas to flow through the fluid passageway **110**, and the outlet port **114** is connected to the collection system **10** (e.g. via the AGSS **30**). The inlet **112** and outlet **114** may have connectors for connecting the conservation valve **22** to piping, tubes, or other mediums for transporting liquids or gases. For example, the inlet **112** may have a ¼-inch NPT female thread. Similarly, the outlet **114** may have a ¼-inch NPT female thread.

(91) The actuator **116** is generally located within the fluid passageway **110** and may be any device for controlling fluid flow, such as an electronically controlled solenoid valve. In some embodiments, the actuator **116** may be a solenoid valve, a part-turn actuator such as a ball valve, or a linear actuator such as a diaphragm valve. In some embodiments, the actuator **116** may be operated by electronics, pneumatics or hydraulics.

(92) The actuator **116** may open and close the fluid passageway **110** based on the presence or absence of exhaust gases coming from the source (e.g. the anaesthetic machine **20** or the patient **18**). As such, the conservation valve **22** may include a sensor **120** that detects the presence or absence of exhaust gases so as to open and close the fluid passageway. The sensor **120** may be connected to the source of exhaust gases upstream of the valve body **108**. For example, in FIG. **1**, the sensor **120** is connected between the patient **18** and the anaesthetic machine **20**.

(93) In some embodiments, the sensor **120** may detect the presence or absence of one or more compounds in the exhaust gases that indicate whether or not there is a flow. For example, the sensor **120** may detect a compound such as oxygen. In particular, the percentage of oxygen in the exhaust flow may indicate whether or not a patient is breathing through the anaesthetic machine **20**, and as such, may indicate whether or not the anaesthetic agent is present in the exhaust gas.

(94) In use, the sensor **120** may measure an ambient oxygen content corresponding to an ambient condition, for example, when the anaesthetic machine **20** is not in use. The ambient oxygen content

may be approximately 21%, which is typically the natural oxygen content in air. When the sensor **120** detects a change in oxygen content, for example, a measured oxygen content above the ambient oxygen (e.g. because the anaesthetic machine is supplying enriched oxygen) or below the ambient oxygen (e.g. the patient is consuming oxygen), this may indicate the anaesthetic machine is in use, and the valve **22** may be opened.

(95) In some embodiments, the sensor **120** may continue monitoring the oxygen content to determine if ambient conditions have re-established such that measured oxygen content returns to approximately the ambient oxygen content, which may indicate the anaesthetic machine **20** is no longer in use. If this occurs, the valve **22** may be closed.

(96) In some embodiments, the valve **22** may open and close based on a measured oxygen content falling within a range around the ambient oxygen content. For example, the range may account for error tolerances in the sensor **120**, such as an error of plus or minus approximately 10%. In some embodiments, the range may be approximately 19% to 23% oxygen content when the ambient oxygen content is 21% and the valve **22** may open when the oxygen content falls outside of the 19% to 23% range. The range may also be selected to account for the amount of oxygen a patient consumes, which may be approximately 5%.

(97) In some embodiments, the sensor **120** may measure the oxygen content after a period of time or over a particular time interval. For example, after opening the valve based on a change in the measured oxygen content, the sensor **120** may measure the oxygen content every sixty seconds, or may measure and calculate the average oxygen content over sixty seconds. This may help account for fluctuations in the oxygen content, which may be associated with error tolerances for the sensor **120**, changes in the patient's breathing, or other factors. In some embodiments, the time period or time interval may be longer or shorter than sixty seconds. For example, in some instances the time period or time interval may be between approximately ten seconds and five minutes.

(98) After the valve **22** opens, the valve **22** may remain open for a particular waiting period even if the measured oxygen content returns to a level corresponding to the ambient oxygen content or an appropriate range. For example, in some instances the waiting period may be any time between approximately one minute and approximately one hour. In some particular embodiments, the waiting period may be approximately one minute.

(99) Providing a waiting period may help reduce the possibility of closing the valve prematurely, for example, when the sensor **120** makes an erroneous measurement.

(100) In some embodiments, the sensor **120** may detect other compounds, such as anaesthetic agents or carbon dioxide. In some embodiments, the sensor **120** may detect a flow or change in pressure as a trigger for opening and closing the valve.

(101) Referring to FIG. **4**, the conservation valve **22** may also include a processor **130** for controlling the actuator **116** based on conditions detected by the sensor **120**. The conservation valve **22** may also include a power supply **132** for providing power to the processor **130**, the sensor **120** and the actuator **116**.

(102) As shown, in some embodiments, one or more subcomponents of the conservation valve **22** may be located inside a housing **140**, which may be made of metal, plastic, or another suitable material. For example, the valve body **108**, the actuator **116**, and the processor **130** may be located inside the housing **140**, while the sensor **120** and the power supply **132** may be located outside the housing **140**. In some embodiments, the sensor **120** and the power supply **132** may also be located inside the housing **140**.

(103) The conservation valve **22** may tend to increase efficiency of the AGSS **30**. For example, if the AGSS **30** operates continuously, it may operate even when the anaesthetic machine **20** is turned off and there are no anaesthetic agents in the flow. This may waste power and reduce efficiency, as the AGSS **30** will be processing gas even when no anaesthetic agents are present.

(104) In contrast, the conservation valve **22** may isolate the source of exhaust gas from the AGSS **30** when the anaesthetic machine **20** is turned off. This may increase the efficiency of the AGSS **30**

because the power source may not have to work as hard when one or more conservation valves **22** are closed.

(105) In some embodiments, the conservation valves may also allow the AGSS **30** to shut off when one or more (e.g. all) of the anaesthetic machines **20** are turned off. This may further increase the efficiency of the AGSS **30**.

(106) The conservation valve **22** may also increase the efficiency of the collection system **10** by isolating the sources of exhaust gas from the collection system **10** when the anaesthetic machine **20** is turned off and there are no anaesthetic agents in the exhaust gas flow.

(107) More particularly, the conservation valve **22** may increase the efficiency of the dryer. For example, when there are no anaesthetic agents in the exhaust gas flow, the dryer **40** may unnecessarily remove moisture. Closing the conservation valve **22** tends to isolate these inactive sources from the dryer **40**, which may increase the potential operation time for the dryer **40** and may reduce the need to purge and regenerate the dryer **40**.

(108) The conservation valve **22** may also increase the efficiency of the collectors **42**. For example, when room air (without anaesthetic agents) flows through collectors **42** that use adsorbent material, the clean air may displace captured anaesthetic agents and carry those anaesthetic agents to the exhaust **46**. This may have the undesired effect of purging the collector **42** of anaesthetic agents, which may reduce the collection efficiency of the collectors **42**. Closing the conservation valves **22** tends to reduce the amount of room air that is processed by the collectors **42** and may prevent unintentional purging.

(109) Referring now to FIG. 5, illustrated therein is a conservation valve **200** connected to an anaesthetic machine **20** according to another embodiment. The conservation valve **200** is similar in many respects to the conservation valve **22**, and where suitable, similar elements are given similar reference numerals incremented by one hundred.

(110) One difference is that the conservation valve **200** includes a flow sensor **220** for measuring the flow of exhaust gas from the anaesthetic machine **20**. The flow sensor **220** may control the operation of the actuator **216** to selectively open and close the fluid passageway **210** based on the presence or absence of fluid flow.

(111) As shown, the flow sensor **220** is located upstream of the valve body **208**. For example, the flow sensor **220** may be connected to the inlet of the anaesthetic machine **20** and may receive exhaust gases from a patient. The flow sensor **220** may measure the flow rate in terms of volume or mass of the exhaust gas. If the flow rate drops below a certain threshold, the flow sensor **220** may trigger the actuator **216** to close. If the flow rate exceeds a certain threshold, the flow sensor **220** may trigger the actuator **216** to open.

(112) The flow sensor **220** may also include a pressure sensor, which may help indicate whether or not the anaesthetic machine **20** is in use when the actuator **216** is in the closed position. If the pressure exceeds a certain threshold, the flow sensor **220** may trigger the actuator **216** to open.

(113) In some embodiments, the flow sensor **220** may communicate with the actuator **216** via the processor **230**. In other embodiments, the flow sensor **220** may communicate directly with the actuator **216**.

(114) In some embodiments, the flow sensor **220** may cooperate with another sensor to determine the presence or absence of exhaust gas flowing from the source. For example, the processor **230** may determine if there is a flow of exhaust gas based on both the flow rate from the flow sensor **220** and the content of a particular compound in the flow as determined by the other sensor (e.g. the sensor **120**). This may be useful when there is ambient flow through the anaesthetic machine **20** but anaesthetic agents are not being delivered to the patient.

(115) The conservation valve **220** may also include a switch **250** (e.g. a flow switch or limit switch) connected to the valve body **208** within the fluid passageway **210** between the inlet **212** and the outlet **214**, and downstream of the actuator **216**. A flow switch helps determine whether the actuator **216** is open or closed based on the presence or absence of a flow after the actuator **216**,

and as such, the flow switch tends to confirm whether the valve **200** is operational. In contrast, a limit switch helps determine whether the actuator **216** is open or closed based on the position of the actuator **216**, and as such, the limit switch tends to confirm whether the valve **200** is operational in a different way compared to the flow switch.

(116) Turning now generally to FIGS. **6** to **10**, in some embodiments it may be desirable that at least some nitrous oxide gas be removed from the exhaust gases before venting the exhaust gases to atmosphere. Nitrous oxide is a greenhouse gas believed to account for a significant proportion of atmospheric heating effects, as well as cause ozone depletion. Furthermore, producing nitrous oxide can be expensive. Accordingly, it may be beneficial if at least some nitrous oxide can be removed and/or collected (e.g. for reuse) as opposed to being vented to atmosphere.

(117) Turning now specifically to FIG. **6**, illustrated therein is a first subsystem **300** of a system for collecting anaesthetic agents, and which is configured to cooperate with one or more other subsystems (as described in FIGS. **7** to **10**) for removing at least some of the nitrous oxide prior to venting to atmosphere.

(118) As shown, the first subsystem **300** includes an AGSS **330** (which may be the same as or similar to the AGSS **30** as generally described above). The AGSS **330** is in fluid communication (e.g. via piping **313**) with one or more sources of exhaust gases, such as one or more operating rooms **312** having one or more patients **318** therein. As shown the piping **313** allows exhaust gases to pass from the operating rooms **312** to the AGSS **330**, including nitrous oxide (N.sub.2O), anaesthetic agents such as halogenated drugs (Hal. Drugs or HD), water or moisture (H.sub.2O), and optionally in some cases air.

(119) In turn, the AGSS **330** is in fluid communication with one or more other subsystems. For example, as indicated by reference numeral “A”, in some embodiments the AGSS **330** may communicate with a second subsystem **302** shown in FIG. **7**. Similarly, as indicated by reference numeral “B”, in some embodiments the AGSS **330** may communicate with a third subsystem **304** shown in FIG. **8**. Furthermore, as indicated by reference numeral “C”, in some embodiments the AGSS **330** may communicate with a fourth subsystem **306** shown in FIG. **9**. Finally, as indicated by reference numeral “D”, in some embodiments the AGSS **330** may communicate with a fifth subsystem **308** shown in FIG. **10**.

(120) In some embodiments, the AGSS **330** may include one or more filters **330a** for filtering the incoming exhaust gases. For example, the filters **330a** could be ultra-low particulate air (ULPA) filters. In other embodiments, the filters **330a** could be high efficiency particulate air (HEPA) filters.

(121) In some embodiments, a Med Vac may be used as the AGSS **330**.

(122) Turning now specifically to FIG. **7**, as shown therein the second subsystem **302** is operable remove at least a portion of the nitrous oxide from the exhaust gases before venting at least some of those gases to atmosphere.

(123) The second subsystem **302** includes a collection system **310**, which may be similar to or the same as the collection system **10** as generally described above, and which is operable to receive gases from the AGSS **330**. For example, the collection system **310** may include components such as a dryer **340** (which could be the same as or similar to the dryer **40** as described above) for removing moisture from the incoming exhaust gases, and one or more collectors **342** located downstream from the dryer **340** for collecting and removing anaesthetic agents from the exhaust gas. In some embodiments, the collectors **342** could be the same as or similar to the collectors **42** as generally described above.

(124) In this embodiment, the collector **342** is shown as an adsorbent collector although other collectors could also be used.

(125) In some embodiments, the collection system **310** may include one or more filters **301a** for filtering the incoming gases, such as a ULPA or HEPA filter. As shown, the filters **301a** may be provided upstream of the dryer **340** and collector **342**.

(126) As shown, moisture removed by the dryer **340** can be vented to atmosphere via an exhaust **346** (coupled to the dryer **340** via a fluid conduit or piping **341**). Furthermore, in this embodiment, the anaesthetic agents collected using the collectors **342** are sent for disposal (e.g. via a fluid conduit or piping **343**). For example, in some embodiments the anaesthetic agents may be collected in large (e.g. 45-gallon) drums and then shipped to a disposal facility.

(127) As shown, in this embodiment, the gas products exhausted by the collectors **342** are sent via piping **345** to one or more reaction chambers **347**. Generally, each reaction chamber **347** is operable to decompose the nitrous oxide (N.sub.2O) into its component parts, namely nitrogen gas (N.sub.2) and oxygen (O.sub.2), according to the following relationship:



(128) For example, the reaction chamber **347** may operate at an elevated temperature to heat the incoming nitrous oxide and encourage the decomposition of the nitrous oxide into nitrogen gas and oxygen. In some embodiments the reaction chamber **347** may be operated at temperatures of around 200 degrees Celsius. In other embodiments, the reaction chamber **347** may be operated at higher or lower temperatures.

(129) In some embodiments, one or more catalysts may be provided to the reaction chamber **347** to facilitate driving the decomposition of the nitrous oxide into nitrogen and oxygen. For example, the catalyst may include precious metals (e.g. platinum, palladium, rhodium, etc.) or metal oxides (e.g. cupric oxide, chromium oxide, ferric oxide, etc.).

(130) By breaking down the nitrous oxide into its component parts, at least some of the nitrous oxide that would otherwise be vented to atmosphere through the exhaust **346** can be converted into other (less damaging) gases. As such, the gases sent via piping **349** from the reaction chamber **347** to the exhaust **346** will tend to include reduced amounts of nitrous oxide (and in some events, may be substantially or even completely stripped of nitrous oxide).

(131) Turning now specifically to FIG. 8, the third subsystem **304** is also operable to work with the first subsystem **300** for removing at least a portion of the nitrous oxide from the exhaust gases. Generally, the third subsystem **304** is substantially similar to the second subsystem **302** as shown in FIG. 7, except that in this embodiment, the anaesthetic agents collected using the one or more collectors **342** are sent for reclamation (e.g. via piping **343** to a reclamation plant) instead of for disposal. In some embodiments the anaesthetic agents may be collected in large (e.g. 45-gallon) drums and then shipped to a reclamation facility.

(132) Turning now specifically to FIG. 9, as shown therein the fourth subsystem **306** is also operable to work with the first subsystem **300** for removing at least a portion of the nitrous oxide from the exhaust gases. The fourth subsystem **306** may include elements similar to the second subsystem **302** and third subsystem **304**, including a collection system **310** having a dryer **340** and one or more collectors **342**.

(133) However, in this embodiment the gases exiting the collectors via piping **345** are sent to a compressor **351** or other similar component, and are not sent to a reaction chamber **347**.

(134) The compressor **351** is operable to compress the gas mixture, which may result in higher concentrations of nitrous oxide. The compressed nitrous oxide gas is then sent via piping **353** to a nitrous storage module **355**, or alternatively (as indicated by reference numeral "E") to a nitrous oxide manifold **359** shown in FIG. 6.

(135) In some embodiments, a Pressure Swing Adsorption (PSA) or other module **352** may be provided downstream of the compressor **351** to help remove nitrogen gas from the nitrous oxide gas stream. In some embodiments, a PSA module **345c** may be provided upstream of the compressor **351** to help remove oxygen gas from the gas stream prior to compression in the compressor **351**.

(136) The nitrous storage module **355** may collect nitrous oxide for subsequent processing off-site so that the nitrous oxide may eventually be reused.

(137) The nitrous oxide manifold **359** may be located on-site at the medical facility (in some

embodiments) and includes one or more storage tanks. The nitrous oxide manifold **359** may be used to supply nitrous oxide (e.g. via one or more pipes **361**) to one or more patients **318** in one or more operating rooms **312** during subsequent medical procedures.

(138) In some embodiments, one or more cleaning or purification steps may be performed at the nitrous oxide manifold **359** to ensure that the nitrous oxide gas collected therein is suitable for subsequent usage (e.g. up to medically acceptable regulatory standards). For example, the purification steps may include removal of carbon dioxide using a carbon dioxide filter (e.g. a soda lime filter).

(139) By collecting and storing the nitrous oxide gases (e.g. either in the manifold **359** or storage module **355**), as opposed to venting the nitrous oxide to atmosphere, undesired pollution tends to be reduced. Furthermore, the nitrous oxide can be reused for subsequent medical procedures, which may result in cost savings.

(140) Turning now specifically to FIG. **10**, as shown therein the fifth subsystem **308** is also operable to work with the first subsystem **300** for removing at least a portion of the nitrous oxide from the exhaust gases. The fifth subsystem **308** may include many elements similar to the second subsystem **302**, third subsystem **304**, and fourth system **306**, such as a collection system **310** having a dryer **340** and one or more collectors **342** (e.g. an adsorbent collector **342** or cryogenic collector **342a**).

(141) Similar to as described above with respect to FIG. **7**, and as shown in FIG. **10**, at least some of the gases exiting the collectors **342** may be fed via piping **345a** to one or more reaction chambers **347** operable to decompose the nitrous oxide into nitrogen gas and oxygen. For example, as shown the reaction chamber **347** may be a combustion chamber that is heated and fed a stream of combustible gas, such as methane gas (CH_4) or another hydrocarbon, to encourage the decomposition of nitrous oxide. In other embodiments, the reaction chamber **347** may include a reaction chamber **347a** having a heated catalyst therein for encouraging the decomposition of the nitrous oxide.

(142) Furthermore, as described above with respect to FIG. **9**, and as shown in FIG. **10**, in some embodiments at least some of the gases exiting the collectors **342** may be fed via piping **345b** to one or more compressors **351** so that they may be compressed for storage for subsequent use. In some embodiments, a PSA module **352** may be provided downstream of the compressor **351** to help remove nitrogen gas from the gas stream.

(143) In some embodiments, after the PSA module **352**, at least some of the gas stream may be sent to the nitrogen manifold **359** shown in FIG. **6**. In other embodiments, at least some of the gas downstream from the PSA module **352** may be sent to a high pressure compressor **351a** for additional compression (e.g. in some examples, to approximately 745 psi, or higher or lower pressures) for storage in a high pressure storage tank **355a**. In some embodiments, the high pressure compressor **351a** may be used on its own without the first compressor **351**.

(144) As shown, in this embodiment the fifth subsystem **308** also includes an oxygen collector **360** for removing oxygen from the gas stream, which may be located upstream of the reaction chamber **347**. It has been discovered that removing the oxygen (e.g. using an oxygen collector **360** as shown) can further facilitate driving the decomposition of nitrous oxide into nitrogen and oxygen in the reaction chambers **347**. Accordingly, extracting oxygen from the gas stream may further facilitate providing exhaust gases with at least some of the nitrous oxide removed.

(145) As shown, the oxygen collector **360** may be positioned downstream of the air dryer **340** and upstream of the one or more collectors **342**. In some embodiments, the oxygen collector **360** could include an adsorbent material for removing the oxygen. In some embodiments, the oxygen collector **360** could include a membrane separator for removing oxygen from the gas stream. In some embodiments, the oxygen collector **360** could include a Pressure Swing Adsorption (PSA) module.

(146) As shown, in some embodiments the oxygen collected by the oxygen collector **360** may be

vented to atmosphere via piping **362**.

(147) In other embodiments, the oxygen can be collected for subsequent use. For example, the oxygen may be sent via piping **364** to a compressor **366** where it may be compressed (e.g. to approximately 50 psi). The compressed oxygen may then be sent (as shown by reference character “F”) to an oxygen manifold **372** shown in FIG. **6**. The oxygen manifold **372** may in turn be used to supply oxygen (e.g. via one or more pipes **374**) to one or more patients **318** in one or more operating rooms **312** during subsequent medical procedures.

(148) In some embodiments, one or more cleaning or purification steps may be performed at the oxygen manifold **372** to ensure that the oxygen gas collected therein is suitable for subsequent usage. For example, the purification steps may include removal of carbon dioxide using a carbon dioxide filter (e.g. a soda lime filter).

(149) In some embodiments, the compressed oxygen may be further compressed in a high pressure compressor **368** (e.g. in some examples to a pressure of around 2200 psi, or higher or lower pressures) and then stored in a high pressure storage tank **370** for subsequent use. In some embodiments, the high pressure compressor **368** may be used on its own without the first compressor **366**.

(150) By collecting and storing the oxygen (e.g. either in the oxygen manifold **372** or storage tank **370**), the oxygen can be reused for subsequent medical procedures, which may result in cost savings.

(151) Referring now to FIGS. **11** and **12**, illustrated therein is a system for collecting anaesthetic agents, including a first subsystem **400** and a second subsystem **402**.

(152) Referring to FIG. **11**, the first subsystem **400** may include elements similar to the first subsystem **300** shown in FIG. **6**, such as the nitrous oxide manifold **359**, the oxygen manifold **372**, and piping **361** and **374**. The manifolds **359** and **373** may be in fluid communication with the second subsystem **402** as generally indicated at reference numerals “H” and “I” respectively, or another subsystem (e.g. subsystem **306** or subsystem **308**).

(153) The first subsystem **400** also includes an AGSS **430** (e.g. similar to the AGSS **30** or the AGSS **330**) in fluid communication via piping **413** with one or more sources of exhaust gases, such as one or more operating rooms **412** having one or more patients **418** therein. The exhaust gases may include nitrous oxide (N.sub.2O), anaesthetic agents such as halogenated drugs (Hal. Drugs or HD), water or moisture (H.sub.2O), biological substances (BIO), and in some cases air.

(154) One difference is that the first subsystem **400** includes at least one filter (e.g. filters **431a**, **431b**, and/or **431c**) between the patient **418** and the AGSS **330** for filtering incoming exhaust gases, for example, to remove particulates such as the biological substances. In particular, the patient **418** might expel biological substances, which might be entrained within the exhaust gases. If left unfiltered, these biological substances might contaminate the overall system and may interfere with system operation, including the collection of anaesthetic agents and/or extraction of nitrous oxide. Furthermore, health regulations might prohibit the subsequent sale or use of collected anaesthetic agents if there are biological substances in the collected anaesthetic agents.

(155) As shown in FIG. **11**, the subsystem **400** includes one filter **431a** within the operating room **412** (e.g. proximal to a mask on the patient), and a bank of two filters **431b** and **431c** outside the operating room **412**.

(156) Providing filters inside the operating room **412** (e.g. filter **431a**) permits early filtration of exhaust gases, which might improve system operation (e.g. collection efficiency). Furthermore, providing filters inside the operating room **412** might also enable medical staff to replace filters within the operating room **412**, for example, so that a new filter may be used for each particular patient.

(157) Providing filters outside the operating room **412** (e.g. filters **431b** and **431c**) might allow centralized filtration of exhaust gases, which might reduce operating costs, and might increase filtration efficiency. Furthermore, providing a bank of at least two filters fluidly coupled together in

parallel (e.g. filters **431b** and **431c**) may provide system redundancy, and may allow one filter (e.g. filter **431a**) to be operational while the other filter (e.g. filter **431b**) is being replaced.

(158) While the filters **431a**, **431b** and **431c** are shown in FIG. **11** as having particular locations, in other embodiments, the filters may have other locations. For example, some filters (e.g. filters **430a** and **430b**) may be located within or after the AGSS **430**. The subsystem **400** may also include more or less filters than shown.

(159) The filters **431a**, **431b**, and **431c** may be the same, similar, or different types of filters. For example, one or more of the filters may be ultra-low particulate air (ULPA) filters, and one or more of the filters may be high efficiency particulate air (HEPA) filters.

(160) The filters of the first subsystem **400** are also arranged in a plurality of stages, including, a first stage filter (e.g. filter **431a**) followed by a second stage filter (e.g. the bank of filters **431b** and **431c**) downstream of the first stage filter. Providing two or more stages of filters may improve filtration of exhaust gases. For example, the first stage (e.g. filter **431a**) may include a first type of filter (e.g. a HEPA filter), which may remove certain types of substances (e.g. a first type of particulate or biological substance), and the second stage (e.g. filters **431b** and **431c**) may include a second type of filter (e.g. a ULPA filter), which may remove other substances (e.g. a second type of particulate or biological substance).

(161) Another difference between the subsystem **400** and the subsystem **300** is that the AGSS **430** includes a bank of at least two filters **430a** and **430b** (e.g. similar to the bank of filters **431a** and **431b**). This allows one filter (e.g. filter **430a**) to be operational while the other filter (e.g. filter **430b**) is being replaced, for example.

(162) Furthermore, each filter in the AGSS **430** may be connected to a respective power source **433a** and **433b** (e.g. a vacuum pump, blower or fan). Each power source may be turned on when the respective filter is installed and operational, and may be turned off when the respective filter is inoperative (e.g. being replaced).

(163) In some embodiments, the subsystem may only include one bank of filters (e.g. the bank of filters **430a** and **430b** in the AGSS **430**).

(164) The outlet of the AGSS **430** is in fluid communication with one or more other subsystems. For example, as indicated by reference numeral “G”, in some embodiments the AGSS **430** may communicate with a second subsystem **402** shown in FIG. **12**. In some embodiments, the AGSS may be in fluid communication with other subsystems, such as the subsystems **302**, **304**, **306** and **308** described previously.

(165) Referring now to FIG. **12**, the second subsystem **402** generally receives exhaust gases from the first subsystem **400** as indicated by the reference numeral “G”. In other embodiments, the second subsystem **402** may receive exhaust gases from other subsystems (e.g. the subsystem **300**).

(166) The second subsystem **402** may include many elements similar to the second subsystem **302**, third subsystem **304**, fourth subsystem **306**, and fifth subsystem **308**, such as a collection system **410** (e.g. similar to the collection systems **10** and **310**), one or more reaction chambers **347** (e.g. a combustion chamber **347** or a catalytic reaction chamber **347a**) for decomposing nitrous oxide, one or more compressors (e.g. compressors **351** or **351a** for compressing nitrous oxide, or compressors **366** or **368** for compressing oxygen), one or more PSA modules **352** (e.g. for removing nitrogen gas), and one or more storage tanks (e.g. a nitrous oxide storage tank **355a** or an oxygen storage tank **370**). The second subsystem **402** may also supply extracted nitrogen and/or oxygen to the first subsystem **400**, or another subsystem (e.g. the subsystem **300**), as generally indicated at reference numerals “H” and “I”.

(167) As indicated above, the collection system **410** may include many elements similar to the collection systems **10** and **310**, such as a dryer **440**, one or more collectors **442** (e.g. an adsorbent collector **442** and/or cryogenic collector **442a**), and an oxygen collector **460**.

(168) One difference between the subsystem **402** and the subsystem **302** is that the collection system **410** also includes a condenser **480** and a disinfection module **482** located upstream of the

collectors **442**.

(169) The condenser **480** removes moisture from the exhaust gases, and outputs a dehumidified exhaust gas. The condenser **480** may expel the extracted moisture (e.g. through a drain **481**). The condenser **480** may be a heat-exchanger having a coolant flowing therethrough for cooling the exhaust gases so as to condense and extract the moisture. In some embodiments, the condenser **480** may be configured to extract a substantial portion of the moisture, for example, so that the dryer **440** has substantially less moisture to extract. Extracting the moisture from the exhaust gases may also improve efficiency of the disinfection module **482** as will be described below.

(170) The disinfection module **482** disinfects (e.g. sterilizes) the exhaust gases by eliminating one or more life forms such as microorganisms (e.g. bacteria, viruses, and fungi) and outputs a disinfected exhaust gas (e.g. to the dryer **440**). More particularly, the disinfection module **482** may eliminate life forms that remain in the exhaust gas after the exhaust gas has been filtered by the upstream filters (e.g. filters **430a**, **430b**, **431a**, **431b**, or **431c** of the first subsystem **400**). For example, the upstream filters may remove larger biological substances (e.g. larger spores, paramecium, bacteria, viruses, etc), but might not remove smaller biological substances (e.g. small bacteria and viruses). Accordingly, the disinfection module **482** may be configured to eliminate these smaller biological substances.

(171) In some embodiments, the disinfection module **482** may utilize a non-contact and non-chemical treatment such as ultraviolet (UV) radiation. More particularly, the disinfection module **482** may utilize UV radiation having a particular wavelength and a power level selected based on the types of biological substances (e.g. pathogens) that pass through the upstream filters (e.g. filters **430a**, **430b**, **431a**, **431b**, or **431c**). For example, the upstream filters may be ULPA filters that are 99.999% effective at removing contaminants larger than 120 nanometers, and the disinfection module **482** may apply radiation at a wavelength and power level so as to be 99.9% effective at eliminating biological substances smaller than 120 nanometers. As an example, the disinfection module may apply UV-C radiation in the range of about 240-280 nanometers and at a power level above about 30,000 microwatts-second per square centimeter ($\mu\text{w-s/cm}^2$), for example, using a low-pressure mercury vapour lamp that produces UV radiation at a wavelength of 254 nanometers.

(172) The disinfection module **482** may also include a UV intensity monitor, for example, to indicate when UV lamps need to be changed.

(173) As shown in FIGS. **11** and **12**, the UV radiation disinfection module **482** is located downstream of one or more filters (e.g. e.g. filters **430a**, **430b**, **431a**, **431b**, or **431c**), which may improve the effectiveness of the UV radiation disinfection module **482**. In particular, the upstream filters may remove larger contaminants, which might otherwise shield UV radiation and prevent the radiation from eliminating smaller biological substances.

(174) In some embodiments, the disinfection module **482** may use other disinfection techniques, such as other forms of radiation (e.g. gamma radiation), heat disinfection, and/or chemical treatment (e.g. ozone, chlorine, or hydrogen peroxide).

(175) As shown in FIG. **12**, the disinfection module **482** is located downstream of the condenser **480**. As such, the disinfection module **482** receives a dehumidified exhaust gas from the condenser **480**. Accordingly, the condenser **480** can remove a substantial amount of moisture prior to disinfecting the exhaust gas, which may increase the efficiency or effectiveness of the disinfection module **482**. For example, some life forms travel in moisture, and it might be possible to remove some of these life forms by extracting moisture from the exhaust gases. Once the moisture has been extracted, the disinfection module **482** may have fewer life forms to eliminate.

(176) In other embodiments, the condenser **480** may be located downstream of the disinfection module **482**. This configuration offers the ability to disinfect the exhaust gas flow prior to condensation, which may be desirable, for example, where regulations require treatment of any extracted moisture prior to being discharged to a sewage system (e.g. via the drain **481**).

(177) In some embodiments, the subsystem **402** may include another disinfection module located on the drain **481**, which may be configured specifically for water treatment so as to reduce or eliminate contaminants flowing down the drain **481**.

(178) While some exemplary locations for the condenser **480** and disinfection module **482** have been described, the condenser **480** and disinfection module **482** may have other locations within the subsystem. For example, the condenser **480** and disinfection module **482** may be located within the operating room **412**, which may allow disinfection/sterilization of exhaust gases during localized collection of anaesthetic agents (e.g. without a centralized collection system).

(179) Another difference between the subsystem **402** and the subsystem **302** is that the collection system **410** includes a dryer regeneration module **490** for regenerating the dryer **440** (e.g. when the dryer includes desiccant that is full of moisture). For example, the dryer regeneration module **490** may supply dry air to purge moisture from the dryer **440** and vent the moisture through an exhaust (e.g. via piping **441**). As shown, the dryer regeneration module **490** may include a heated blower purge module **490a** (which may be the same as or similar to the dryer regeneration module **60**). The dryer regeneration module **490** may also include a regeneration dryer **490b** for removing moisture from the main dryer **440** as described below.

(180) The regeneration dryer **490b** may include a twin tower desiccant dryer having a first desiccant tower **491a**, a second desiccant tower **491b**, and a compressor **491c** for blowing air from the regeneration dryer **490** through the main dryer **440**. Each of the desiccant towers **491a** and **491b** is selectively operable in a drying mode for regenerating the main dryer **440**, or a purging mode so as to purge moisture from the desiccant tower.

(181) In operation, the compressor pulls outside air through a first desiccant tower **491a**, which is operating in drying mode so as to remove moisture from the outside air. The compressor then blows the dehumidified outside air (e.g. purging air) through the main dryer **440** so as to regenerate the desiccant therein. While the first desiccant tower **491a** is in drying mode, the second desiccant tower **491b** is in purge mode such that a portion of the dehumidified purging air exiting the first desiccant tower **491a** flows through the second desiccant tower **491b** so as to purge moisture from the second desiccant tower **491b**.

(182) When the first desiccant tower **491a** is full of moisture, the operation switches so that the second desiccant tower **491b** operates in drying mode while the first desiccant tower **491a** operates in purge mode. As such, the twin tower desiccant dryer allows continuous purging of the main dryer **440**, even when one desiccant tower is saturated with moisture. The system may include control valves for controlling when the desiccant towers **491a** and **491b** are in drying mode and purge mode.

(183) The main dryer **440** may also include a plurality of dryers fluidly coupled in parallel (e.g. similar to the twin tower desiccant dryer of the regeneration dryer **490b**). For example, a first dryer (e.g. a first desiccant tower **440a**) may remove moisture from the exhaust gases while the regeneration module **490** (e.g. the regeneration dryer **490b**) regenerates or purges moisture from a second dryer (e.g. a second desiccant tower **440b**).

(184) Then, when the first desiccant tower **440a** is saturated with moisture, the operation may switch such that the regeneration module **490** begins regenerating the first desiccant tower **440a** while the second desiccant tower **440b** removes moisture from the exhaust gases. The system may include control valves for controlling when the two desiccant towers **440a** and **440b** are in operation (i.e. removing moisture from the exhaust gases) or being purged by the regeneration module **490**.

(185) In some embodiments, air supplied from the regeneration module **490** (e.g. the regeneration dryer **490b**) may also be used to operate one or more pneumatically controlled valves within the system (e.g. valves **50**, **70**, **72**, **84**, **86**, **88** described previously with respect to the collection system **10**). More particularly, the compressor **491c** of the regeneration dryer **490b** may supply pressurized air for operating the control valves.

(186) The collection system **410** may also include a bank of filters (e.g. filters **492a** and **492b**) between the dryer **440** and the collectors **442**. This bank of filters offers another opportunity to remove contaminants and other particulates from the exhaust gases prior to entering the collectors **442**. For example, the filters **492a** and **492b** may remove particulates such as desiccant particles entrained in the exhaust gases after passing through the dryer **440**. The filters **492a** and **492b** might also remove biological substances that remain in the exhaust gases after being treated by the disinfection module **482**. In particular, the filters **492a** and **492b** may filter out microorganisms that were killed by the disinfection module **482**.

(187) In some embodiments, the filters **492a** and **492b** may be located in other positions, for example, upstream of the oxygen collector **460** or upstream of the dryer **440**.

(188) Referring now to FIGS. **13** and **14**, illustrated therein is another system for collecting anaesthetic agents, including a first subsystem **500** and a second subsystem **502**.

(189) Referring to FIG. **13**, the first subsystem **500** is similar in many respects to the first subsystem **400** shown in FIG. **11**. One difference is that the filters **431b** and **431c** are located downstream of the AGSS **430**. Positioning the filters after the AGSS **430** may increase filtration efficiency, for example, because the exhaust gases leaving the AGSS **430** are under positive pressure, and the filters may be more efficient when filtering exhaust gases under positive pressure (e.g. approximately 3 psi) as opposed filtration under negative pressure upstream of the AGSS (e.g. 12-20 inches-Hg).

(190) Another difference is that the subsystem **500** identifies carbon dioxide (CO₂) as a substance within the exhaust gases. It will be understood that previous embodiments may also include carbon dioxide within the exhaust gases even if not expressly identified.

(191) Referring now to FIG. **14**, the second subsystem **502** is similar in many respects to the second subsystem **402** shown in FIG. **12**. One difference is that the disinfection module **482** is located further downstream within the collection system **410** (i.e. between the filters **492a** and **492b** and the collectors **442**). Positioning the disinfection module **482** further downstream may increase the effectiveness of the disinfection module **482**, for example, because other system components might remove substances that would otherwise interfere with the disinfection process. In particular, particles such as water molecules might block UV radiation that is used to disinfect the exhaust gases. Accordingly, it might be desirable to remove these substances (e.g. using dryers, filters, membrane separators and the like) prior to disinfecting the exhaust gas.

(192) Another difference is that the second subsystem **502** includes a number of devices and processes for drying the exhaust gases prior to collection. For example, as described previously with respect to FIG. **12**, at least some of the exhaust gases may be fed to the dryer **440** (e.g. via piping **520a**) for removing moisture.

(193) Furthermore, in some embodiments, at least some of the exhaust gases may be fed to a compressor **530** (e.g. via piping **520b**) for increasing the pressure of the exhaust gases. For example, the compressor **530** may receive the exhaust gases at a low pressure (e.g. at approximately 3 psi) and may output the exhaust gases at a high pressure (e.g. at approximately 80-120 psi). The compressor **530** may be a variable speed compressor (e.g. as shown in FIG. **14**) or another type of compressor such as a fixed speed compressor.

(194) The high pressure exhaust gases leaving the compressor **530** may be fed to one or more dryers. For example, at least some of the high pressure exhaust gases may be fed to a desiccant dryer **540** (e.g. similar to the dryers **40**, **340**, and **440**). The dryer **540** may be more efficient at removing moisture from high pressure exhaust gases in comparison to low pressure exhaust gases, for example, because the flow rate of the exhaust gases through the dryer **540** can be increased, or because the size of the dryer **540** can be reduced, or both.

(195) Furthermore, at least some of the high pressure exhaust gases leaving the compressor **530** may be fed to a membrane dryer **542**. The membrane dryer **542** generally includes a porous membrane separator that removes water vapor from the high pressure exhaust gases, for example,

based on selective adsorption of water molecules. Similar to the desiccant dryer **540**, the membrane dryer **542** may be more efficient at removing moisture from high pressure exhaust gases in comparison to low pressure exhaust gases.

(196) As shown, the portion of the exhaust gases leaving the dryer **440** and the desiccant dryer **540** may pass through a drug concentrator **560** and then through a set of filters **492a** and **492b**. The filters **492a** and **492b** may filter out desiccant particles that may be entrained in the exhaust gases after passing through the dryers **440** and **540**.

(197) The drug concentrator **560** may remove substances from the exhaust gases (e.g. oxygen, carbon dioxide and the like) so as to increase the concentration of anaesthetic agents prior to collection using the collectors **442**. In some embodiments, the drug concentrator **560** may include an adsorbent material, a membrane separator (e.g. similar to the oxygen collector **360**), or a PSA module. The substances collected by the drug concentrator **560** may be vented to atmosphere (e.g. via piping **462**), or may be collected for subsequent use (e.g. oxygen may be routed to the compressor **366** via piping **364** for storage in the storage tank **370**, or for resupply to patients via the oxygen manifold **372**).

(198) While FIG. **14** refers to a few specific examples of drug concentrators, the drug concentrator **560** is more generally any device that removes a substance from the exhaust gas prior to collection of the anaesthetic agents (e.g. to increase the concentration of the anaesthetic agent within the exhaust gas). Accordingly, in some embodiments, the drug concentrator **560** may include filters (e.g. filters **430a**, **430b**, **431a**, **431b**, **431c**, **492a**, **492b**, etc), dryers (e.g. dryers **440**, **540**, **542**, or the condenser **480**), membrane separators, PSA modules, and the like.

(199) Similar to the exhaust gases leaving the dryer **440** and the desiccant dryer **540**, the portion of the exhaust gases leaving the member dryer **542** may also flow through a drug concentrator **562**, which may be the same as, or similar to, the drug concentrator **560**. In some embodiments, the membrane dryer **542** and the drug concentrator **562** may be incorporated into a single membrane separator that removes both water molecules and other gases such as oxygen and carbon dioxide.

(200) While the exhaust gases leaving the membrane dryer **542** pass through a drug concentrator, it can be seen that these exhaust gases do not pass through any additional filters (e.g. like the filters **492a** and **492b**). The filters may be omitted, for example, because the membrane dryer **542** does not use desiccant particles that might otherwise contaminate the exhaust gases (whereas the dryers **440** and **540** might introduce desiccant particles and it might be desirable to filter out those desiccant particles).

(201) In some embodiments, the condenser **480** may be located after the compressor **530**, which may increase the efficiency of the condenser **480**. For example, the condenser **480** may be more efficient at removing moisture from high pressure exhaust gases (approximately 80-120 psi) in comparison to low pressure exhaust gases (e.g. approximately 3 psi) because the water vapor in the high pressure exhaust gases has a higher condensation temperature (i.e. a higher dew point). A higher condensation temperature allows the condenser **480** to operate at a higher temperature (e.g. without the need to refrigerate the exhaust gases), which may increase the efficiency of the condenser **480**.

(202) Another difference between the subsystem **502** and the subsystem **402** is that the gas products exhausted by the collectors **442** may be sent via piping **345a** to another type of reaction chamber **347b** for decomposing nitrous oxide within the gas products. In particular, the reaction chamber **347b** may include a plasma arc conversion module that feeds the gas products through a high temperature plasma torch so as to decompose the nitrous oxide into nitrogen and oxygen.

(203) While some of the embodiments described above may refer to collecting or reclaiming halogenated drugs, the apparatus, systems and methods described herein may generally be used for the collection of various types of anaesthetic agents, including halogenated drugs and other agents.

(204) Although the above description provides examples of one or more methods, systems and

apparatuses, it will be appreciated that other methods, systems and apparatuses may be within the scope of the present description as interpreted by one of skill in the art.

Claims

1. A system for collecting an anaesthetic agent, the system comprising: at least one anaesthetic gas scavenging system (AGSS) for receiving exhaust gas from a plurality of sources, the exhaust gas including the anaesthetic agent to be collected; and a central collection system for receiving the exhaust gas, the central collection system comprising: a disinfection module for disinfecting the exhaust gases and outputting a disinfected exhaust gas; and at least one collector for receiving the disinfected exhaust gas, wherein the at least one collector is configured to adsorb the anaesthetic agent from the exhaust gas.
 2. The system of claim 1, wherein the disinfection module utilizes a non-contact and non-chemical treatment for disinfecting the exhaust gas.
 3. The system of claim 2, wherein the disinfection module utilizes ultraviolet radiation.
 4. The system of claim 1, further comprising at least one filter upstream of the disinfection module for filtering the exhaust gas.
 5. The system of claim 4, wherein the at least one filter includes a bank of filters fluidly coupled in parallel.
 6. The system of claim 4, wherein the at least one filter includes a first stage filter, and a second stage filter downstream of the first stage filter.
 7. The system of claim 4, wherein the disinfection module utilizes ultraviolet radiation.
 8. The system of claim 7, wherein the ultraviolet radiation has a wavelength and a power level selected to eliminate a substantial number of biological substances remaining in the exhaust gas after being filtered by the at least one filter.
 9. The system of claim 8, wherein the at least one filter is configured to remove substances larger than 120 nanometers, and wherein the disinfection module is configured to apply UV-C radiation such that the wavelength has a range of between about 240 nanometers and about 280 nanometers, and the power level is above about 30,000 microwatts-second per square centimeter.
 10. The system of claim 1, further comprising a condenser for removing moisture from the exhaust gas.
 11. The system of claim 10, wherein the disinfection module is located downstream of the condenser.
 12. A system for collecting an anaesthetic agent, the system comprising: a central collection system for receiving exhaust gas from a plurality of sources, the exhaust gas including the anaesthetic agent to be collected, the central collection system comprising: a disinfection module for disinfecting the exhaust gases and outputting a disinfected exhaust gas; and at least one collector for receiving the disinfected exhaust gas, wherein the at least one collector is configured to collect the anaesthetic agent from the exhaust gas.
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